

Hierarchical Institutions for Contract Invalidity

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Abstract. Contracts can be modelled from a rights-first perspective as bundles of legal rights, which naturally supports contrary-to-duty (CTD) reasoning: when primary rights are not fulfilled, secondary rights and remedies arise. CTD captures the normative dynamics *within* a contract. In legal practice, however, admissible remedies also depend on an institutional determination *about* the contract — whether it is *valid*, *void*, or *voidable*. In this paper, we introduce a two-level architecture for contract governance, called *hierarchical institutions*, with two coupled state machines: an object-level contract-instance machine for drafting, acceptance, performance, revision, and escalation; and a meta-level institutional machine that, once a dispute is escalated, classifies the contract, may determine avoidance for voidable contracts, and dispatches an admissible repair family. This separation clarifies that the same dispute can be routed to enforcement or compensation when the contract is treated as valid, and to restoration such as restitution or unwinding when the contract is void or avoided. We sketch an agentic reasoning model in which agents update beliefs and intentions under institutional rules to anticipate these outcomes. Two scenarios inspired by Hungarian private law illustrate the framework: usury leading to voidness and gross disparity leading to voidability with avoidance.

1 Introduction

Contracts, understood as bundles of rights, provide a natural point of interaction between agents and institutions. Agents draft and negotiate the rights embodied in a contract, while institutions give legal effect to those rights. Between these points, a contract evolves as a structured artefact that can be performed, violated, repaired, or even deprived of legal effect. In AI&Law, this dynamic is often studied through *contrary-to-duty* (CTD) reasoning: if a primary right in the contract is not fulfilled, which secondary rights become operative, who may trigger them, and how the system returns to a stable normative state.

CTD is well-suited for rights-first reasoning *within* an effective contract [8]. It captures how contractual rights guide performance and how breaches trigger remedy structures. However, a focus on CTD alone can leave a crucial dimension underexplored: reasoning *about* the contract as an institutional procedure. In legal practice, parties do not merely anticipate what happens *if* a right is

violated; they also anticipate what happens *if* the agreement itself is challenged as producing no (or only defeasible) contractual legal effects. This institutional “shadow” influences drafting, negotiation, and escalation decisions, and is therefore central to agreement technologies and agentic systems operating under legal rights [15].

A contract may be *valid*, *void* (no contractual legal effect), or *voidable* (effective unless and until successfully challenged, typically with retroactive consequences). This kind of validity control is not just another CTD branch inside the contract. It is a meta-level institutional operation—performed by courts, regulators, or other legal authorities—that determines whether contractual effects are legally operative and enforceable [13]. When invalidity “blocks” a contract, it blocks contractual normative effects and enforceability, not the parties’ ability to act in the world. Parties may already have transferred money, goods, or services; in such cases the legal system may route consequences through restoration mechanisms (e.g., restitution or unwinding) rather than through contractual enforcement or damages.

Institutions are structurally present throughout contractual interaction as background constraints: legal norms shape what agents should do and what they expect others to do. Institutional procedures typically become *procedurally active* when parties face an issue (e.g., a dispute or a validity challenge) and seek authoritative repair. In this paper we focus on this dispute-triggered pathway, while keeping the architecture compatible with additional institutional checkpoints.

Our motivating case study is institutional determination of validity outcomes that cannot be decided “inside” the contract. We illustrate this with two running examples framed by Hungarian private law: usury (formed but classified as void) and gross disparity (formed and classified as voidable; may become retroactively ineffective after successful avoidance).

The contrast can be summarised as follows:

- **CTD / violation reasoning** addresses what happens when agents do not fulfil what the contract requires (e.g., late delivery, non-payment). In many agreement technologies, the handling of such disruption is partly specified by the parties in the contract itself (e.g., damages clauses, revision mechanisms, escalation paths) and partly supplied by default legal remedies [15].
- **Voidness / voidability reasoning** addresses what happens when the legal system does not determine the legal effects of a contract (or part of it), or determines it as defeasible and later terminated retroactively. Here, the handling is primarily imposed by institutional doctrine: denying enforceability, enabling avoidance procedures, and imposing restoration duties such as restitution or unwinding.

This difference matters for agentic systems: reasoning only in CTD terms can treat every disruption as repairable within the contractual framework, while institutional classification (void/voidable) can deny enforceability from the outset or unwind the relationship after performance.

In this paper, our research questions are:

1. How do contractual violations differ from institutional invalidity outcomes and their consequences?
2. How can a legal institution determine an accepted but violated agreement as valid/void/voidable and route repair accordingly, including avoidance and restoration?
3. How can agents anticipate institutional classification and repair when deciding whether to accept, perform, revise, escalate, or comply under a contract?

Our approach is design-oriented and operational. Rather than proposing a new logic, we develop an explicit institutional model that separates what happens *within* a contract from what happens *about* the contract at the institutional level. We model this separation using a hierarchical transition system with two coupled layers: an object-level procedure capturing drafting, acceptance, performance, revision, and dispute escalation, and a meta-level procedure capturing institutional classification of validity, voidness, and voidability, including avoidance and retroactive effects. Within this setting, repair is treated as a central organising principle: both violation and invalidity are understood as perturbations that require restoration, but they differ in trigger, authority, and routing of admissible mechanisms. On the agent side, we provide a lightweight practical reasoning model that allows agents to anticipate institutional classifications and repair outcomes, based on belief and intention update under institutional rules.

The paper is structured as follows. Section 2 introduces the running examples. Section 3 presents the hierarchical institutional architecture, distinguishing object-level contractual operations from meta-level validity and avoidance control, and formalises the corresponding state machines and repair pathways. Section 4 develops the agentic reasoning model and illustrates how agents reason about violation, voidness, and institutional repair when deciding how to act under a contract. Section 5 discusses related work, and Section 6 concludes with directions for future research.

2 Case Study and Motivating Examples

Legal systems classify contracts that are *void* from contracts that are *voidable*, alongside the default case of *valid* contracts. Voidability is typically associated with an avoidance mechanism: an entitled party may attack enforceability through legally prescribed procedure (e.g., notice to the counterparty or court action); if avoidance succeeds, the contract becomes retroactively ineffective and restoration duties may follow (e.g., restitution/unwinding). We use this pattern in the Hungarian Civil Code (Act V of 2013, sections 6:88-6:98) to illustrate why validity control belongs *above* the contract. We use two scenarios to classify two distinct institutional determination patterns:

- **Usury:** an agreement is formed but institutionally determined as *void*.
- **Gross disparity:** an agreement is formed and institutionally determined as *voidable*, i.e., effective unless and until successful avoidance leads to retroactive voidness and restitution/unwinding.

Symbol	Informal meaning
$\text{Void}(k)$	contract k is void (no contractual legal effect)
$\text{Voidable}(k)$	contract k is voidable (effective but defeasible)
$\text{Valid}(k)$	contract k is valid (binding and enforceable)

Table 1. Meta-level validity statuses used in the motivating examples.

Example U: Usurious loan offer (void) A lender and a borrower conclude an agreement k_U whose surrounding circumstances meet a legal ground for usury, represented as $\text{Usurious}(k_U)$. Institutionally, this ground can support classification $\text{Void}(k_U)$: the agreement is treated as producing no enforceable contractual effects, and the system should not proceed as if ordinary breach remedies (enforcement/compensation) were available under the contract. At the agent level, the parties may anticipate this risk already during negotiation and treat it as a reason to reject, revise, or avoid escalation.

Example D: Sale with gross disparity (voidable + avoidance) A seller and buyer conclude an agreement k_D whose circumstances meet a legal ground for gross disparity without gift intention, represented as $\text{GrossDisparity}(k_D)$ and $\text{NotGratuitous}(k_D)$. Institutionally, these grounds can support classification $\text{Voidable}(k_D)$: the contract is effective but defeasible, because an entitled party may exercise an avoidance mechanism that, if successful, leads to retroactive ineffectiveness and triggers restitution/unwinding. At the agent level, this defeasibility affects acceptance and performance incentives, including whether parties attempt private revision or escalate to dispute.

These two examples motivate a hierarchical separation: *object*-level contractual operations (formation and performance) do not by themselves determine a contract’s institutional standing; instead, validity outcomes (Valid , Voidable , Void) and their repair consequences must be controlled at a *meta*-level that can block enforcement, enable avoidance, and impose restoration.

3 Hierarchical Institutions

The institutional functions of *contract classification* and *contract repair* constitute the foundation of our architecture of *hierarchical institutions*. Unlike CTD reasoning *within* an effective contract, these functions are performed by legal institutions (courts, regulators, arbitral bodies) and therefore concern authoritative assessments *about* the contract as an institutional object.

Contract classification determines a contract instance’s legal status as one of Valid , Voidable , or Void . Contract repair determines which *families* of legally admissible consequences can be ordered to resolve a dispute arising from non-fulfilment or contestation of an accepted contract. Based on these functions, we introduce a two-level architecture of contract governance, called a *hierarchical institution*, coupling two *state machines*: (1) an *object-level* contract-instance

machine modelling the parties' interaction, and (2) a *meta-level* institutional machine governing validity review, (optional) avoidance classification, and repair dispatch.

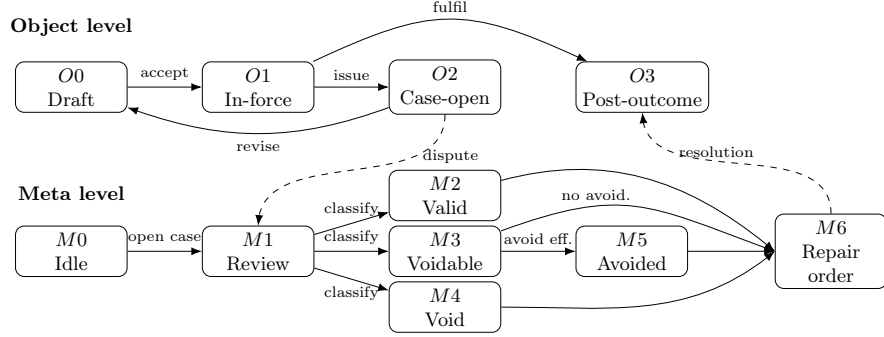


Fig. 1. Compact coupled governance machine: object-level execution and meta-level validity classification and repair control. Dashed arrows denote coupling (dispute escalation and resolution feedback).

3.1 A Two-Lane Hierarchy

Institutions are typically present as *background constraints* on contracting, but they become *procedurally active* once a dispute arises and a case is opened. The two layers of the hierarchy are shown in Figure 1. The object-level lane describes how a contract progresses through formation, performance, disruption handling, and post-outcome steps. The meta-level lane captures how a legal institution determines the contract's institutional status. Depending on that determination, the institution may deny enforceability (**Void**), treat the contract as defeasible and enable an avoidance pathway (**Voidable**), or confirm its enforceability (**Valid**). The coupling between lanes is represented through classificatory *counts-as* norms [9]: institutional determinations make certain facts and acts count as institutional statuses (e.g. **Voidable**) and admissible repair families (e.g. **Restitution**), thereby constraining which downstream actions are institutionally admissible for the parties.

We model each contract instance k as evolving in a two-lane transition system for the hierarchy, presented in Table 2. A run interleaves (i) object-level moves by the contractual parties and (ii) meta-level moves by the institution. To make the machine auditable, we distinguish (1) *fact transitions* (logged events such as non-fulfilment, evidence surfacing, and benefit transfers) from (2) *act transitions* (exercises of powers such as acceptance, dispute filing, avoidance, classification, and remedy orders). A governance configuration is a pair $\langle o, m \rangle$ where $o \in \{O0, O1, O2, O3\}$ is the current object-level state and $m \in \{M0, \dots, M6\}$ is the current meta-level state.

Two transitions explicitly couple the lanes at the level of global configurations: escalation activates the institution, and the institutional order feeds back to object-level execution. Concretely, escalation moves the global configuration from $\langle O2, M0 \rangle$ to $\langle O2, M1 \rangle$ (opening institutional review), while the dispatched order moves the configuration from $\langle O2, M6 \rangle$ to $\langle O3, M0 \rangle$ (execution/closure).

3.2 Counts-as Coupling and Repair Routing

The machine becomes operational once we specify how *grounds* count as institutional statuses, and how these statuses constrain admissible repair families. In institutional review ($M1$), the institution establishes *grounds*: legally relevant conditions supported by evidence and doctrine (e.g. $\text{Usurious}(k)$, $\text{GrossDisparity}(k)$, $\text{NotGratuitous}(k)$). These grounds serve as inputs to classificatory counts-as norms producing a validity status.

Using the motivating examples in Section 2, three key counts-as rules for classification are:

$$\begin{aligned} \text{Formed}(k) &\Rightarrow_{\text{ca}} \text{Valid}(k) && (C_{\text{valid}}) \\ \text{Formed}(k) \wedge \text{Usurious}(k) &\Rightarrow_{\text{ca}} \text{Void}(k) && (C_U) \\ \text{Formed}(k) \wedge \text{GrossDisparity}(k) \wedge \text{NotGratuitous}(k) &\Rightarrow_{\text{ca}} \text{Voidable}(k) && (C_D) \end{aligned}$$

Rule (C_U) (voidness, $M1 \rightarrow M4$) captures Example U; rule (C_D) (voidability, $M1 \rightarrow M3$) captures Example D; and (C_{valid}) represents the default classification for formed contracts in the absence of established defeating grounds. Note that $\text{Valid}(k)$, $\text{Void}(k)$, $\text{Voidable}(k)$ are mutually exclusive. A logic of counts-as norms must capture their logical relations to *preserve consistency*; these can be formalised as follows.

$$\begin{aligned} \text{Valid}(k) &\Rightarrow \neg \text{Voidable}(k) && (Ex_1) \\ \text{Valid}(k) &\Rightarrow \neg \text{Void}(k) && (Ex_2) \\ \text{Voidable}(k) &\Rightarrow \neg \text{Void}(k) && (Ex_3) \end{aligned}$$

where $\Rightarrow$ is the classical implication. In other words, we understand mutual exclusion in the sense of classical logic.

The statuses $\text{Valid}(k)$, $\text{Voidable}(k)$, and $\text{Void}(k)$ are treated as mutually exclusive outputs of institutional classification. In particular, when multiple classification rules could apply, the institutional machine produces a single status upon leaving $M1$ (e.g. by a fixed priority among applicable grounds).

Voidability enables an avoidance pathway ($M3 \rightarrow M5$), whereas non-avoidance proceeds directly to repair dispatch ($M3 \rightarrow M6$). In the latter case, the entitled contractual party does not exercise the power of avoidance, and the institution recognises that the contract remains valid and that repair should therefore be undertaken. $\text{AvoidAttempt}(i, j, k)$ represents an avoidance exercise by party i toward j (e.g. a legally relevant notice or claim). In our abstract model, $\text{AvoidAttempt}(i, j, k)$ is recorded at the meta level only when the institution determines it as an effective exercise for contract instance k in the given case.

ID	Lane	Name	What it means	Enabled moves	outputs /
<i>O0</i>	obj	Draft/Negotiate	No formed instance yet (or parties are re-negotiating).	Propose terms, revise, reject; may reach acceptance.	
<i>O1</i>	obj	In-force / Perform	Contract is formed and treated as operating at the object level.	Perform/monitor; may reach fulfilment or detect a perturbation.	
<i>O2</i>	obj	Perturbation / Case-open	A disruption is detected (non-fulfilment, disagreement, or validity challenge).	Choice point: private revision vs. escalation; (if escalated) await meta-level output.	
<i>O3</i>	obj	Post-outcome	After fulfilment or after institutional resolution; execute termination/repair plan.	Execute ordered repair (e.g. compensation or restitution), close, or continue cooperation.	
<i>M0</i>	meta	Idle (background only)	No active proceeding; only background constraints apply.	On escalation, open institutional review.	
<i>M1</i>	meta	Review (proceeding open)	Institution gathers/assesses grounds and routes the case.	Classify Valid/Voidable/Void; may recognise avoidance; dispatch repair.	
<i>M2</i>	meta	Classified Valid	Contracts are recognised as enforceable.	Breach-handling admissible: enforcement/compensation (if Violated).	
<i>M3</i>	meta	Classified Voidable	Contracts are valid but defeasible; avoidance power is potential for entitled party.	Either (i) dispatch repair while still valid, or (ii) transition to avoided.	
<i>M4</i>	meta	Classified Void	Contractual validities are denied (as an non-enforceable contract).	Restoration admissible: restitution/unwinding (where benefits transferred).	
<i>M5</i>	meta	Avoided (effective avoidance)	Voidability is realised: contract becomes retroactively invalid.	Rescission + restitution/unwinding admissible.	
<i>M6</i>	meta	Repair dispatched / Order	Institution issues the admissible repair family for the case.	Feeds back to object level: parties execute repair/termination accordingly.	

Table 2. Governance states (object lane *O* and meta lane *M*).

$$\text{Voidable}(k) \wedge \text{AvoidAttempt}(i, j, k) \Rightarrow_{ca} \text{AvoidEffective}(k) \quad (C_A)$$

From $\rightarrow$ To	Trigger (guard $\wedge$ act)	Effect (informal)
$O0 \rightarrow O1$	act: acceptance $\text{Accept}(\cdot, \cdot, k)$	(e.g. Instance becomes formed/in-force ($\text{Formed}(k)$, $\text{InForce}(k)$)).
$O1 \rightarrow O3$	fact: fulfilment succeeds	Discharge at object level; move to post-outcome governance.
$O1 \rightarrow O2$	fact: perturbation detected (non-fulfilment, disagreement, validity challenge)	Open a case at object level (choice point: revise vs dispute).
$O2 \rightarrow O0$	act: private revision agreed	Loop back to negotiation without institutional procedure.
$O2 \rightarrow M1$ (coupling)	act: escalation / dispute filing	Institution becomes procedurally active (opens review).
$M1 \rightarrow M2$	counts-as classification (no defeating ground established)	Institution recognises enforceability as valid.
$M1 \rightarrow M3$	counts-as classification (voidability grounds established)	Institution marks contract defensible; avoidance becomes relevant.
$M1 \rightarrow M4$	counts-as classification (voidness grounds established)	Institution denies contractual enforceability; shift to restoration.
$M3 \rightarrow M5$	act: avoidance attempted + institution recognises effectiveness	Retroactive ineffectiveness ($\text{NoEffect}(k)$); rescission/restoration dominates.
$M3 \rightarrow M6$	act: repair routing/order while still not avoided	Dispatch admissible repair while the contract remains valid (e.g. enforce/comp if Violated).
$M2/M4/M5 \rightarrow M6$	act: repair routing/order issued	Institution dispatches the admissible repair family for the case.
$M6 \rightarrow O3$ (coupling)	fact: institutional resolution communicated/entered	Feed back: parties execute repair/termination in post-outcome state.

Table 3. Transitions of hierarchical governance machine (facts vs. acts + coupling).

Intuitively, $\text{AvoidEffective}(k)$ means that the contract becomes retroactively ineffective and the governance pathway shifts to restoration-oriented consequences.

A voidable contract *remains valid unless the power of avoidance is exercised*. This non-avoided situation differs from that of contracts are valid and non-voidable ($\text{Valid}(k)$). It is represented as $\text{Voidable}(k) \wedge \neg \text{AvoidEffective}(k)$. We

thus route breach-driven repair under both **Valid** and (non-avoided) **Voidable**:

$$\begin{aligned}
& (\text{Valid}(k) \vee (\text{Voidable}(k) \wedge \neg \text{AvoidEffective}(k))) \\
& \quad \wedge \text{Violated}(k) \Rightarrow_{\text{ca}} \text{Route}(k, \text{Enforce/Comp}) \\
& \hspace{15em} (R_{\text{eff}}) \\
& \text{Void}(k) \Rightarrow_{\text{ca}} \text{Route}(k, \text{Restitute/Unwind}) \\
& \hspace{15em} (R_0) \\
& \text{AvoidEffective}(k) \Rightarrow_{\text{ca}} \text{Route}(k, \text{Rescind} + \text{Restitute}) \\
& \hspace{15em} (R_A)
\end{aligned}$$

The transition from $M6$ back to the post-outcome state $O3$ is a second coupling, running from the meta level to the object level: the institutional order constrains what the parties execute in $O3$.

Trace sketches (Example U and D).

- **U (usury $\rightarrow$ void):** $O0 \rightarrow O1 \rightarrow O2 \dashrightarrow M1 \rightarrow M4 \rightarrow M6 \dashrightarrow O3$ (route: retribute/unwind).
- **D (gross disparity $\rightarrow$ voidable $\rightarrow$ avoided):** $O0 \rightarrow O1 \rightarrow O2 \dashrightarrow M1 \rightarrow M3 \rightarrow M5 \rightarrow M6 \dashrightarrow O3$ (route: rescind+restitute).

3.3 Repair Control as a Meta-level Submachine

Once a contractual interaction enters a perturbed state, repair control is precisely the meta-level segment $M1 \rightarrow (M2/M3/M4/M5) \rightarrow M6$: the process from review/classification, (optional) avoidance determination, to repair dispatch. See Figure 2 for the detailed architecture.

1. *Perturbation Detector* (supports $O1 \rightarrow O2$): monitors operational and structural anomaly signals.
2. *Institutional Gateway* (supports $O2 \rightarrow M1$ and $M1 \rightarrow M2/M3/M4$): routes the case to the appropriate institutional track.
3. *Repair Dispatcher* (supports $M2/M3/M4/M5 \rightarrow M6$): selects an admissible repair family consistent with validity/avoidance outcomes.
4. *Equilibrium Validator*: checks whether the ordered measures restore an acceptable normative balance under institutional constraints.

4 Agentic Reasoning Model

When agents draft and negotiate a contract, they must take the meta level contract governance into account — to anticipate possible post-outcomes ($O3$) that fulfill the contractual purpose, as well as the potential repairs ($M6$) when the contract is violated.

We propose a formal representation of practical reasoning that enables agents to *reason within* and *about* contracts, with particular emphasis on the coupling process between the object and meta levels. The model integrates two complementary frameworks in order to bridge the gap between institutional rules and individual practical reasoning:

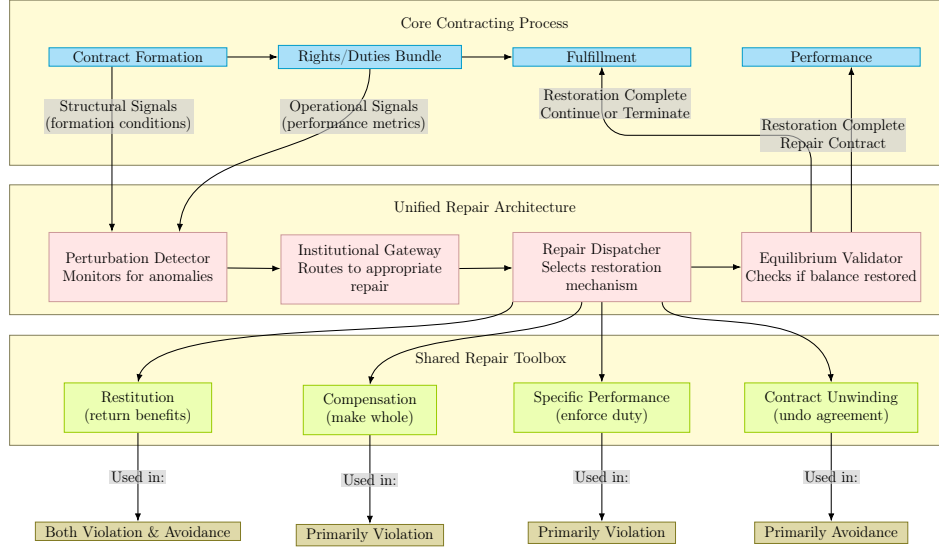


Fig. 2. Repair control as a meta-level submachine that classifies and dispatches admissible repair families.

- Classificatory counts-as norms [9]: These are employed to represent the institutional hierarchy — encompassing both operational procedures and repair controls — grounded in a rights-first perspective [8];
- Agent-based databases: These provide a computational structure to reason about the evolution of legal rights and duties over time [6].

By extending the database approach with counts-as norms, we enable institutional processes of validity classification and their associated repair mechanisms to be analyzed in a systematic, step-by-step operational manner. This synthesis allows agents to reason about their concrete reactions in terms of counts-as norms within their belief and intention databases. In particular, when a contract is counted as activated, breached, or avoided, the framework determines which beliefs and intentions are practically appropriate to adopt.

4.1 The Agentic Machine

We define an *agentic machine* A_i^t (for an agent $i \in \mathbf{Ag}$ at time t) as a structure of three different databases: $\langle \mathcal{B}_i^t, \mathcal{I}_i^t, \mathcal{N}_i^t \rangle$. This model facilitates agent-level practical reasoning regarding the institutional hierarchy.

Beliefs $\mathcal{B}_i^t$. This database is partitioned into strong and weak beliefs: $\mathcal{B}_i^t = \mathcal{B}_{i,s}^t \cup \mathcal{B}_{i,w}^t$. Strong beliefs support *requiring* (hard) gates — those inputs are assessed by institutions such as whether a contract is accepted or unfair, therefore primary for agents’ databases as unchangeable when updating intention; weak beliefs motivate caution and information-seeking — which are derived beliefs from strong beliefs within other databases, such as that a contract is voidable.

This mirrors the previous work [6] in which the belief base is separated from “hard constraints” to “soft evidence”.

Intentions $\mathcal{I}_i^t$. An intention base is consistent with explicit timestamps, storing all consistent *individual* and legal actions for agent i [6]. It is possible for the intention base $\mathcal{I}_i^t$ of agent i to store other agents’ possible actions [16] like avoidance exercising. We adopt the intention-based revision methods that have been well discussed in the existing work [16, 6].

Normative and institutional model $\mathcal{N}_i$. This contains the institution’s *meta-rules* involved in the institutional hierarchy. These are represented as *counts-as norms*. In our loan example, the key norms include counts-as rules (C_{valid}), (C_D), and (C_A). Agents use them to predict institutional outputs. We do not support that the contractual bounded parties share the same model of the institutional rules and so it is possible that $\mathcal{N}_i \neq \mathcal{N}_j$ when $i \neq j$.

4.2 Practical Reasoning and Logical Closure

Agents’ autonomy is governed by the normative boundaries of the institution — the “rules of the game” that structure interaction [14]. Consequently, beliefs are influenced by the social context.

In this model, strong beliefs represent evidence-based truths and facts, while weak beliefs aim at social acceptance [14]. Weak beliefs are derived not only from an agent’s internal states but also from the institutional status in which the agent is embedded. When an agent functions as a “group member” (a party to a contract), these weak beliefs constitute public facts that other members are expected to share.

Logical Closure. Our inference mechanism adopts a Searlean methodology [4] regarding social acceptance. While belief data is information-based, the institutional model is rule-based:

$$\text{weak belief} = \text{closure}_{\text{counts-as}}(\text{strong belief} \cup \text{intention}).$$

More precisely, weak beliefs are derived from strong beliefs and intention, logically closed under the counts-as norms. We do not specify which logic of counts-as norms $\Rightarrow_{\text{ca}}$ to use, but many are available in the literature [9]. When using $Cn_{\text{ca}}(\Gamma)$ to mean the logical closure of Γ on the specific logic of $\Rightarrow_{\text{ca}}$, and so the weak belief database $\mathcal{B}_{i,w}^t$ is defined as $Cn_{\text{ca}}(\mathcal{B}_{i,s}^t \cup \mathcal{I}_i^t \cup \mathcal{N}_i)$. We use $\mathcal{B}_{i,w}^t \models \varphi$ to indicate that $\varphi \in \mathcal{B}_{i,w}^t$. Consider the example of unfair load offer in Section 2. Let the agent machine $A_B^0 = \langle \mathcal{B}_B^0, \mathcal{I}_B^0, \mathcal{N}_B \rangle$ for agent B at time 0 be

- $\mathcal{B}_B^0 = \{\text{Formed}(k_D)\}_B^0$;
- $\mathcal{I}_B^0 = \{\epsilon\}_B^0$ where ϵ indicate the empty intention;
- $\mathcal{N}_B = \{C_{\text{valid}}, C_D, C_A\}$.

Thus, $\mathcal{B}_{B,w}^0 \models \text{Valid}(k_D)$. Without further information, the weak belief database cannot infer any result beyond this and therefore determines the contract to be valid at this stage. Accordingly, the rule (C_{valid}) is applicable, whereas neither (C_D) nor (C_A) applies.

An agentic machine A_i^t is coherent, when $\mathcal{B}_{i,w}^t \not\models \perp$. As $\mathcal{B}_{B,w}^0 \not\models \perp$, the agent machine A_B^0 for agent B at time 0 is coherent.

4.3 Revision Mechanisms

These revision mechanisms are employed to maintain consistency when updating beliefs and intentions [16, 6]. The postulates of revision ensure that when new information conflicts with existing information, the latter is removed or revised accordingly. For example, since **Valid** and **Voidable** are mutually exclusive, if **Valid** is present in the database and a revision requires the addition of **Voidable**, then **Valid** must be removed during the update to preserve consistency.

An intention revision function $\otimes$ maps an agentic machine at time t and an intention to a new agent machine at time $t + 1$ such that

$$\langle \mathcal{B}_i^t, \mathcal{I}_i^t, \mathcal{N}_i \rangle \otimes \alpha = \langle \mathcal{B}_i^{t+1}, \mathcal{I}_i^{t+1}, \mathcal{N}_i \rangle$$

where the following postulates hold:

- $\mathcal{B}_i^t = \mathcal{B}_i^{t+1}$;
- $\langle \mathcal{B}_i^{t+1}, \mathcal{I}_i^{t+1}, \mathcal{N}_i \rangle$ is coherent;
- If $\langle \mathcal{B}_i^{t+1}, \{\alpha\}_i^{t+1}, \mathcal{N}_i \rangle$ is coherent, then $\alpha \in \mathcal{I}_i^{t+1}$;
- If $\langle \mathcal{B}_i^{t+1}, (\mathcal{I}_i^t \cup \{\alpha\})_i^{t+1}, \mathcal{N}_i \rangle$ is coherent, then $(\mathcal{I}_i^t \cup \{\alpha\})_i^{t+1} \subseteq \mathcal{I}_i^{t+1}$;
- $\mathcal{I}_i^{t+1} \subseteq (\mathcal{I}_i^t \cup \{\alpha\})_i^{t+1}$;
- For all $\mathcal{I}_i^{t+1}$ with $\mathcal{I}_i^{t+1} \subset \mathcal{I}_i^{t+1} \subseteq (\mathcal{I}_i^t \cup \{\alpha\})_i^{t+1}$, the machine $\langle \mathcal{B}_i^{t+1}, \mathcal{I}_i^{t+1}, \mathcal{N}_i \rangle$ is not coherent.

We can update the prior agentic machine A_B^0 with the intention of sending a note of avoidance **AvoidAttempt**(L, B, k_D), leading to $A_B^1 = \langle \mathcal{B}_B^1, \mathcal{I}_B^1, \mathcal{N}_B \rangle$ as

$$- \mathcal{B}_B^1 = \{\text{Formed}(k_D)\}_B^1 \text{ and } \mathcal{I}_B^1 = (\mathcal{I}_B^0 \cup \{\text{AvoidAttempt}(L, B, k_D)\})_B^1.$$

The new agentic machine A_B^1 is coherent and it still has $\mathcal{B}_{B,w}^1 \models \text{Valid}(k_D)$. However, the new intention does not bring to any institutional effect like $\mathcal{B}_{B,w}^1 \models \text{Voidable}(k_D)$, since neither the necessary premise **GrossDisparity**(k_D) nor the premise **NotGratuitous**(k_D) is presented in the agentic machine A_B^1 .

Belief Revision. A belief revision function $\circ$ maps an agentic machine at time t and a strong belief formula φ to a new agent machine at time $t + 1$ such that

$$\langle \mathcal{B}_i^t, \mathcal{I}_i^t, \mathcal{N}_i \rangle \circ \varphi = \langle \mathcal{B}_i^{t+1}, \mathcal{I}_i^{t+1}, \mathcal{N}_i \rangle$$

where

- $\mathcal{B}_i^{t+1}$ is the result of revising $\mathcal{B}_i^t$ with a strong belief φ that satisfies the AGM postulates [1, 5];
- $\mathcal{I}_i^{t+1}$ is the result of revising the new beliefs with the empty intention ϵ so that coherence is restored, i.e. $\langle \mathcal{B}_i^{t+1}, \mathcal{I}_i^{t+1}, \mathcal{N}_i \rangle \otimes \epsilon = \langle \mathcal{B}_i^{t+1}, \mathcal{I}_i^{t+1}, \mathcal{N}_i \rangle$.

Now we update A_B^1 with the strong belief **GrossDisparity**(k_D) and this leads to a new agent machine $A_B^2 = \langle \mathcal{B}_B^2, \mathcal{I}_B^2, \mathcal{N}_B \rangle$ as

$$- \mathcal{B}_B^2 = (\mathcal{B}_B^1 \cup \{\text{GrossDisparity}(k_D)\})_B^2 \text{ and } \mathcal{I}_B^2 = \mathcal{I}_B^1.$$

So A_B^2 is still coherent and we further know $\mathcal{B}_{B,w}^2 \models \text{GrossDisparity}(k_D)$. We continue to update A_B^2 with strong belief **NotGratuitous**(k_D), leading to $A_B^3 = \langle \mathcal{B}_B^3, \mathcal{I}_B^3, \mathcal{N}_B \rangle$ such that $\mathcal{B}_{B,w}^3 \models \text{NotGratuitous}(k_D)$. Given the counts-as norms,

this will lead to two new institutional results: $\mathcal{B}_{B,w}^3 \models \text{Voidable}(k_D)$ by applying rule (C_D) and $\mathcal{B}_{B,w}^3 \models \text{AvoidEffective}(k_D)$ by the rule (C_A) .

Because $\text{Valid}(k_D)$ and $\text{Voidable}(k_D)$ are mutually exclusive, the postulates in belief revision require that the prior validity claim $\text{Valid}(k)$ be removed from the updated belief base $\mathcal{B}_{B,w}^3$.

5 Related Work

Institutional legal effects are often represented with *constitutive* (counts-as) rules linking observable facts and actions to institutional statuses and normative consequences. Jones and Sergot explain how designated acts performed in context generate institutional effects such as competence and status change, thus separating physical behaviour from the institutional layer [11]. Subsequent work develops general patterns for constitutive norms and counts-as conditionals, clarifying how institutional facts (including classifications) are derived and how they constrain later moves [9]. In normative multi-agent systems, constitutive rules also serve as a design tool: agent moves count as legal acts and trigger institutional outputs [2]. Our meta-level component follows this tradition by treating validity classification and avoidance determination as constitutive outputs that govern repair routing and admissible continuations.

Rights-first approaches take Hohfeldian positions (claim/duty, power/liability, immunity/disability) as primitives for representing legal relations and their dynamics [10]. Formal reconstructions clarify how these positions can be represented as conditional legal relations and how their consequences propagate [13]. Work on legal competences highlights that *powers* are state-changing legal acts, not merely physical abilities, which is central to modelling avoidance [7]. Recent rights-first CTD models treat remedies and revisions as transformations over live bundles of directed positions [8]. We extend this stance to validity control, showing how institutional outcomes about the contract constrain the repair families available after a dispute.

Normative multi-agent systems and agreement technologies emphasise separating the computation of legal effects from agents' choices of which actions to take [3, 15]. Input/output logic supports modular architectures in which rule sets determine permitted and obligatory outputs given contextual inputs [12]. Architecture-oriented work also treats normative reasoning as a component in managing dynamic systems and change [16]. Our contribution instantiates these ideas for contracts by coupling an object-level contract-instance procedure with a meta-level institutional procedure for validity classification, avoidance determination, and repair routing.

6 Summary and Future Work

In this paper, we presented a rights-first hierarchical governance model that separates reasoning *within* a contract from reasoning *about* the contract as an

institutional procedure. The architecture couples two state machines: an object-level contract-instance machine for drafting, acceptance, performance, breach and escalation, and a meta-level institutional machine that classifies a contract as valid, voidable and void, optionally classifies avoidance as an exercised power with retroactive consequences, and dispatches a repair family. Coupling is given by classificatory counts-as rules from legally relevant grounds to institutional statuses and routing outputs. On the agent side, we introduced a database-based practical reasoning model with time-stamped strong beliefs and intentions (including others' actions), and weak beliefs obtained by closure under the institutional rules, enabling agents to anticipate classification and repair while revising plans. We illustrated the approach with two Hungarian-law traces: usury leading to voidness and restoration, and gross disparity leading to voidability, effective avoidance, and rescission-based repair.

One future work direction is to combine the state automaton for rights-first contrary to duty breach and repair from [8] with the state machine for validity control, voidness, voidability, and avoidance developed in this paper. It yields a unified automaton that covers the full range of rights based notions, including primary rights, contrary to duty rights, revision and compensation rights, and institutional construction and deconstruction of enforceability.

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